Project Description

The project introduces you to recent developments in operating systems research and how to read academic research papers. The project is open-ended. You choose the specific problem and technology you want to study, such as CPU scheduling and optimization, process and thread scheduling, memory management, networking and security, or parallel and distributed computing.

This is a group project with 1-5 group members.

Project Process:

1. Find a recent research paper published within the last five years (2022 to 2026), e.g., IEEE Xplore (https://www.ieee.org/) or the ACM Digital Library (https://dl.acm.org/), and decide the problem you want to work on.
2. Read the paper carefully until you understand both the method and how the authors evaluated it.
3. Implement the method proposed in the paper. The easiest way is to choose a paper whose authors have published their code on GitHub. You can search for these at https://paperswithcode.com/. Be aware that published code is often incomplete or buggy. You can either debug it or switch to a different paper, so take your own risk if you do this in the last minute.
4. Compare the figures and results you generate with those in the paper, and analyze any differences you find. Your results will probably not match exactly, and explaining why is an important part of this project. It is also possible if your results match exactly as what the paper shows.
5. Submit your presentation slides before 11/6, 11:59 pm. The slides should include sections on background, problem statement, novelty, method, simulation results (showing the original figures alongside your own for comparison), and conclusion. You will present your paper to the class during the last two weeks of the semester. In your presentation, do not paste blocks of text onto your slides and read from them. Use key words and short phrases instead, and explain the details in your own words. Try to use figures and diagrams as they communicate ideas better.
6. Submit your final package before 12/1, 11:59 pm. The package must contain the original paper, your final report (see template), and a video showing how you run your code and generate your results.

Note: 1) Your paper must be related to this course. For example, a paper on AI models or techniques by itself will not be accepted, but a paper that applies AI to an operating systems problem will be. 2) Start your project early, as the simulation can take more time than you think. You can ask for HPC resources if needed. 3) Do not overthink this project. The goal is to understand a paper and share it with your classmates. You are not expected to improve on the paper or produce original research. 4) State clearly in your final report what each group member contributed.